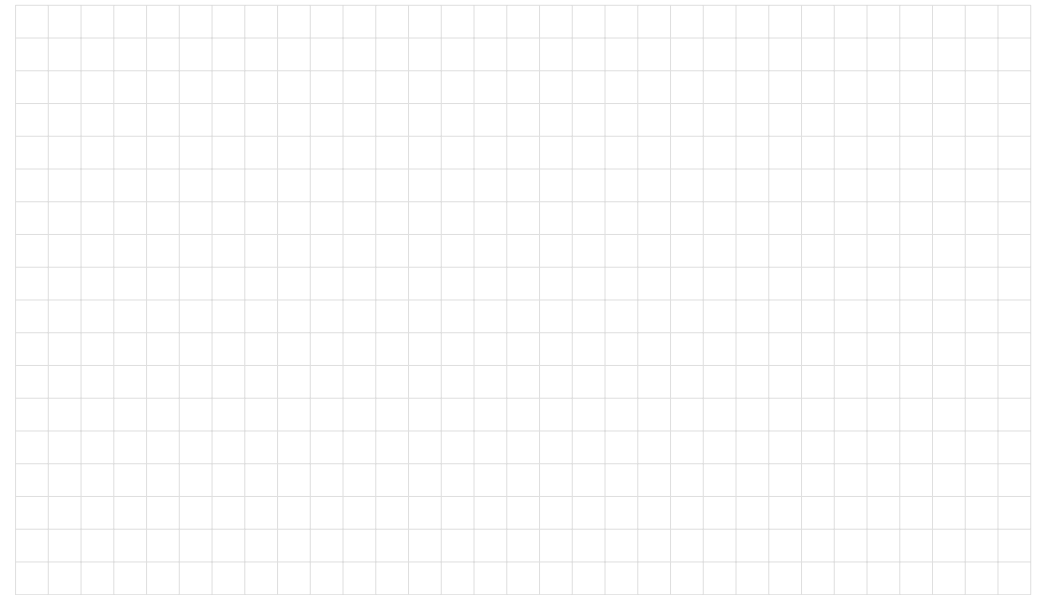


Elements of Graph Theory

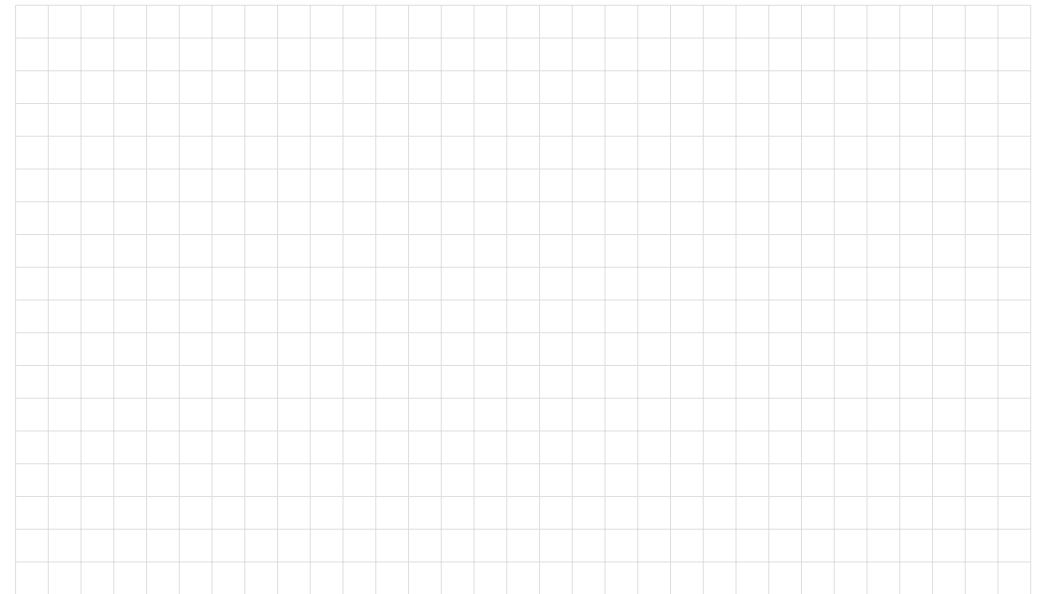
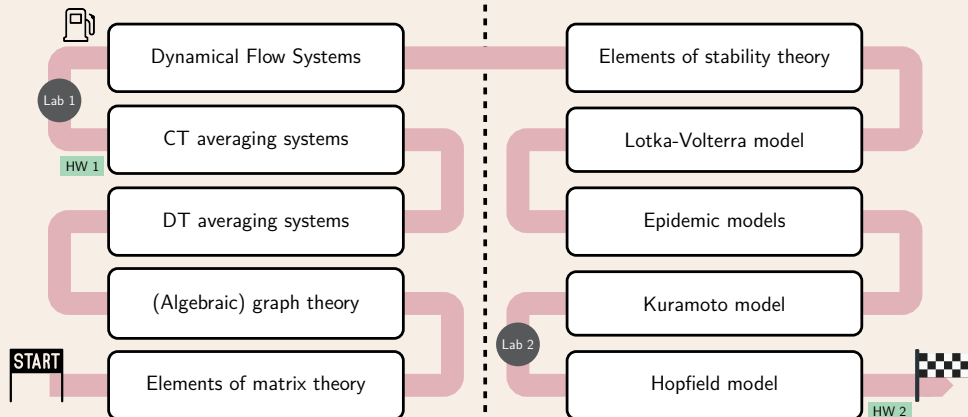
Giacomo Baggio

Network Dynamical Systems
Master Degree in Control Systems Engineering
A.A. 2025-2026



Linear Dynamics

Nonlinear Dynamics



Covered topics

- ▷ Undirected graphs
- ▷ Directed graphs
- ▷ Walks and connectivity in directed / undirected graphs
- ▷ Weighted directed graphs



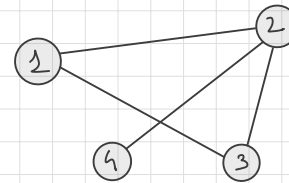
Undirected graphs

Def: An undirected graph (in short, graph) of order n is a pair $G = (V, E)$ where V is a set with n elements, called nodes, and E is a set of unordered pairs of nodes, called (unordered) edges.

- ▷ we denote an unordered edge connecting $v, w \in V, v \neq w$, as $\{v, w\} \in E$
- ▷ if $\{v, w\} \in E$, v and w are neighbors; $\mathcal{N}(v)$ = set of neighbors of v
- ▷ degree of $v \in V$, $d(v)$ = number of neighbors of v
- ▷ $G = (V, E)$ regular if all nodes in V have the same degree

Note: Self-loops and multi-edges are not allowed!

$\{1, 2\}$ $\{ \dots, \{1, 2\}, \{1, 2\}, \dots \}$



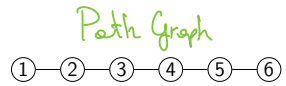
$$G = (V, E)$$

$$V = \{1, 2, 3, 4\}$$

$$E = \{ \underbrace{\{1, 2\}}_{=\{2, 1\}}, \{1, 3\}, \{2, 3\}, \{2, 4\} \}$$

Undirected graphs

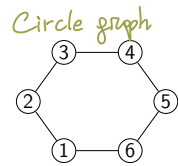
Q: $N(3)$? $d(1)$? Is the graph regular?



$$N(3) = \{2, 4\}$$

$$d(1) = 2$$

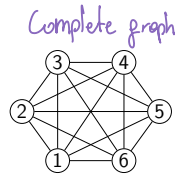
Not regular



$$N(3) = \{2, 4\}$$

$$d(1) = 2$$

Regular



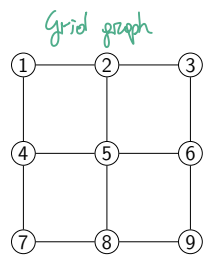
$$N(3) = \{2, 4, 5, 6\}$$

$$d(1) = 5$$

Regular

Undirected graphs

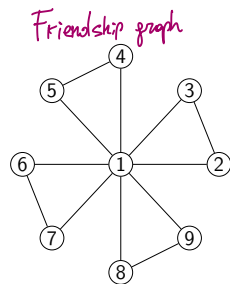
Q: $N(3)$? $d(1)$? Is the graph regular?



$$N(3) = \{2, 6\}$$

$$d(1) = 2$$

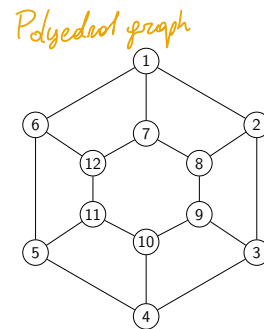
Not regular



$$N(3) = \{2, 10\}$$

$$d(1) = 8$$

Not regular



$$N(3) = \{2, 4, 11\}$$

$$d(1) = 3$$

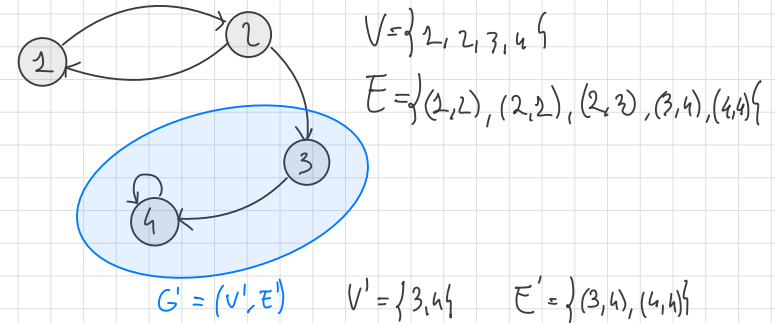
Regular

Friendship graph: Every pair of distinct nodes have a common neighbor

Directed graphs

Def: A directed graph (in short, digraph) of order n is a pair $G = (V, E)$ where V is a set with n elements, called nodes, and $E \subseteq V \times V$ is a set of ordered pairs of nodes, called (ordered) edges.

- ▷ we denote an ordered edge from $v \in V$ to $w \in V$ as $(v, w) \in E$
- ▷ if $(w, v) \in E$ anytime $(v, w) \in E$ then G is undirected
- ▷ an edge from a node v to itself, $(v, v) \in E$, is called self-loop
- ▷ digraph $G' = (V', E')$ is a subgraph of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$



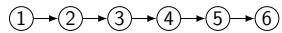
Directed graphs

Def: A directed graph (in short, digraph) of order n is a pair $G = (V, E)$ where V is a set with n elements, called nodes, and $E \subseteq V \times V$ is a set of ordered pairs of nodes, called (ordered) edges.

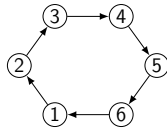
- ▷ if $(v, w) \in E$, v is an in-neighbor of w and w is an out-neighbor of v
- ▷ $\mathcal{N}^{\text{in}}(v)$ ($\mathcal{N}^{\text{out}}(v)$) = set of in-neighbors (out-neighbors) of v
- ▷ in- (out-) degree of $v \in V$, $d_{\text{in}}(v)$ ($d_{\text{out}}(v)$) = number of in- (out-) neighbors of v
- ▷ $v \in V$ source (sink) if $d_{\text{in}}(v) = 0$ ($d_{\text{out}}(v) = 0$)
- ▷ G topologically balanced if $d_{\text{in}}(v) = d_{\text{out}}(v)$ for all $v \in V$

Directed graphs

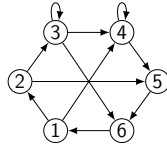
Q: $\mathcal{N}^{\text{in}}(3)$? $\mathcal{N}^{\text{out}}(3)$? $d_{\text{in}}(1)$? Topologically balanced?



$\mathcal{N}^{\text{in}}(3) = \{2\}$
 $\mathcal{N}^{\text{out}}(3) = \{4\}$
 $d_{\text{in}}(1) = 0 \Rightarrow 1$ is a source
 Not balanced



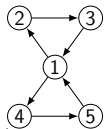
$\mathcal{N}^{\text{in}}(3) = \{2, 4\}$
 $\mathcal{N}^{\text{out}}(3) = \{4, 6\}$
 $d_{\text{in}}(1) = 1$
 Balanced



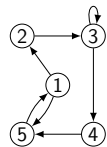
$\mathcal{N}^{\text{in}}(3) = \{2, 3\}$
 $\mathcal{N}^{\text{out}}(3) = \{3, 4, 6\}$
 $d_{\text{in}}(1) = 2$
 Not balanced

Directed graphs

Q: $\mathcal{N}^{\text{in}}(3)$? $\mathcal{N}^{\text{out}}(3)$? $d_{\text{in}}(1)$? Topologically balanced?

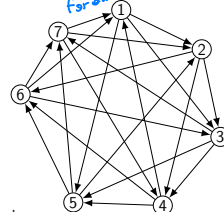


$\mathcal{N}^{\text{in}}(3) = \{2, 3\}$
 $\mathcal{N}^{\text{out}}(3) = \{1, 4\}$
 $d_{\text{in}}(1) = 2$
 Balanced



$\mathcal{N}^{\text{in}}(3) = \{2, 3\}$
 $\mathcal{N}^{\text{out}}(3) = \{3, 4\}$
 $d_{\text{in}}(1) = 2$
 Not balanced

Note: There is a common in-neighbor
 and a common out-neighbor
 for all nodes



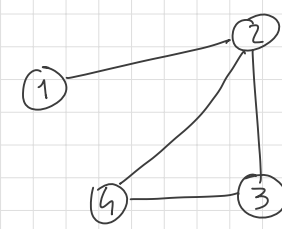
$\mathcal{N}^{\text{in}}(3) = \{1, 2, 4\}$
 $\mathcal{N}^{\text{out}}(3) = \{4, 5, 7\}$
 $d_{\text{in}}(1) = 3$
 Balanced

Tricky! $d_{\text{in}}(1) = d_{\text{out}}(1) = 2$
 $d_{\text{in}}(v) = d_{\text{out}}(v) = 1 \forall v \neq 1$

Walks and connectivity in graphs

Def: A walk (or path) in an undirected graph $G = (V, E)$ is an ordered sequence of nodes such that any pair of consecutive nodes in the sequence is an edge of the graph.

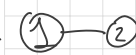
- ▷ simple walk = walk with no repeated nodes, except possibly the first and last
- ▷ cycle = simple walk that starts and ends at the same node → must contain at least 3 nodes
- ▷ G is **connected** if there exists a walk between any two nodes
- ▷ G is **acyclic** if it contains no cycles
- ▷ G is a **tree** if it is connected and acyclic



$(1, 2, 3, 4) \rightarrow \text{Length } 3$

$(1, 2, 4) \rightarrow \text{Length } 2$

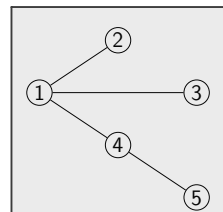
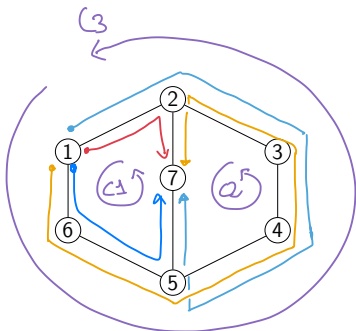
$(1, 2, 3, 4, 2) \rightarrow \text{Length } 4$



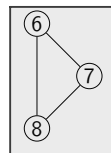
$(1, 2, 2)$ could be a cycle but there are 3 nodes required

Walks and connectivity in graphs

Q: Find a walk from 1 to 7. Is it simple? Any cycle? Connected? Tree?



Walk 1 → 7

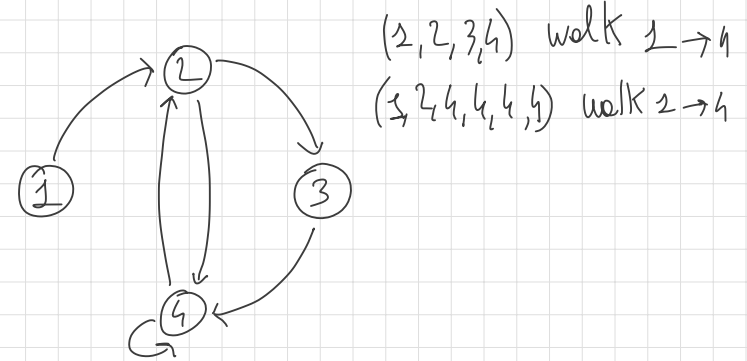


$C_1 = (1, 2, 7, 1)$ $C_2 = (2, 3, 4, 5, 6, 2)$ $C_3 = (1, 2, 3, 4, 5, 6, 1)$

Walks and connectivity in digraphs

Def: A directed walk in a directed graph $G = (V, E)$ is an ordered sequence of nodes such that any pair of consecutive nodes in the sequence is an edge of the digraph.

- ▷ simple (directed) walk = walk with no repeated nodes, except possibly the first/last
- ▷ (directed) cycle = walk that starts and ends at the same node
- ▷ simple (directed) cycle = simple walk that starts and ends at the same node
- ▷ G is **acyclic** if it contains no cycles
- ▷ G is a **directed tree** if it is acyclic and has a node (root) from which every other node is reached by a unique walk



Walks and connectivity in digraphs

Q: Acyclic? Directed tree?



Walks and connectivity in digraphs

Def: A directed graph $G = (V, E)$ is

- (i) **strongly connected** if there is a directed walk from any node to any other node;
- (ii) **weakly connected** if the undirected version of the graph is connected.

Moreover, G has a

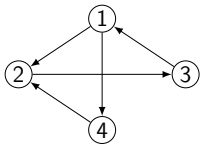
- (iii) **globally reachable node** if one of its nodes can be reached from any other node through a directed walk;
- (iv) **directed spanning tree** if one of its nodes is the root of directed walks to every other node.

Note: 1) G strongly connected $\Leftrightarrow G$ weakly connected

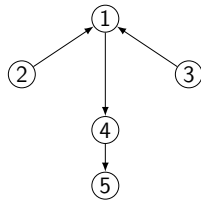
2) G strongly connected $\Leftrightarrow$ every node of G is the root of a directed spanning tree

Walks and connectivity in digraphs

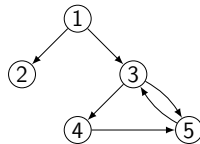
Q: Strongly/weakly connected?



strongly connected



not strongly connected
weakly connected



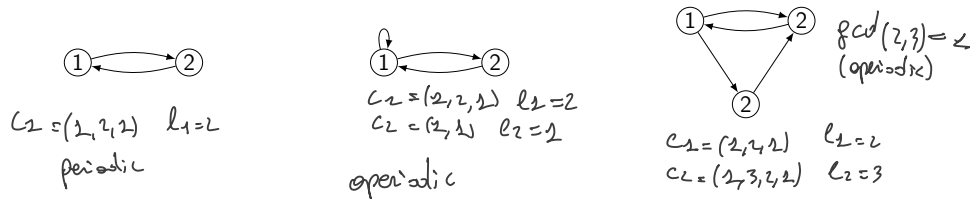
not strongly connected
weakly connected

Walks and connectivity in digraphs

Def: A strongly connected digraph $G = (V, E)$ is periodic if there exists an integer $k > 1$ (the period) that divides the length of every simple cycle of the digraph.

- ▷ G periodic $\iff \gcd\{\text{lengths of simple cycles}\} > 1$
- ▷ G strongly connected digraph with a self-loop $\implies G$ aperiodic

Q: Is G periodic?

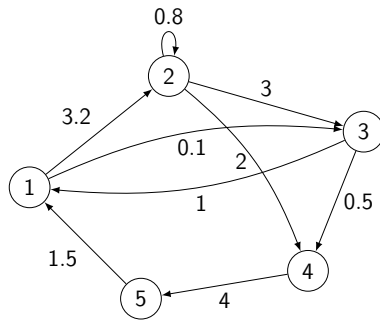


Weighted directed graphs

Def: A directed weighted graph (in short, weighted digraph) of order n is a triplet $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ where (V, E) is a digraph and $\{a_{ij}\}_{(i,j) \in E}$ is collection of weights for the edges E .

- ▷ digraph = weighted digraph with all $a_{ij} = 1$
- ▷ (undirected) graph = weighted digraph with $a_{ij} = 1$ and $a_{ij} = a_{ji}$ for all i, j
- ▷ weighted out-degree of $i \in V$, $d_{\text{out}}(i) = \sum_{j=1}^n a_{ij}$
- ▷ weighted in-degree of $i \in V$, $d_{\text{in}}(i) = \sum_{j=1}^n a_{ji}$
- ▷ G weight-balanced if $d_{\text{out}}(v) = d_{\text{in}}(v)$ for all $v \in V$

Weighted directed graphs



$$a_{12} = 3.2, a_{13} = 0.1$$

$$a_{22} = 0.8, a_{23} = 3, a_{24} = 2$$

$$a_{31} = 1, a_{34} = 0.5$$

$$a_{45} = 4$$

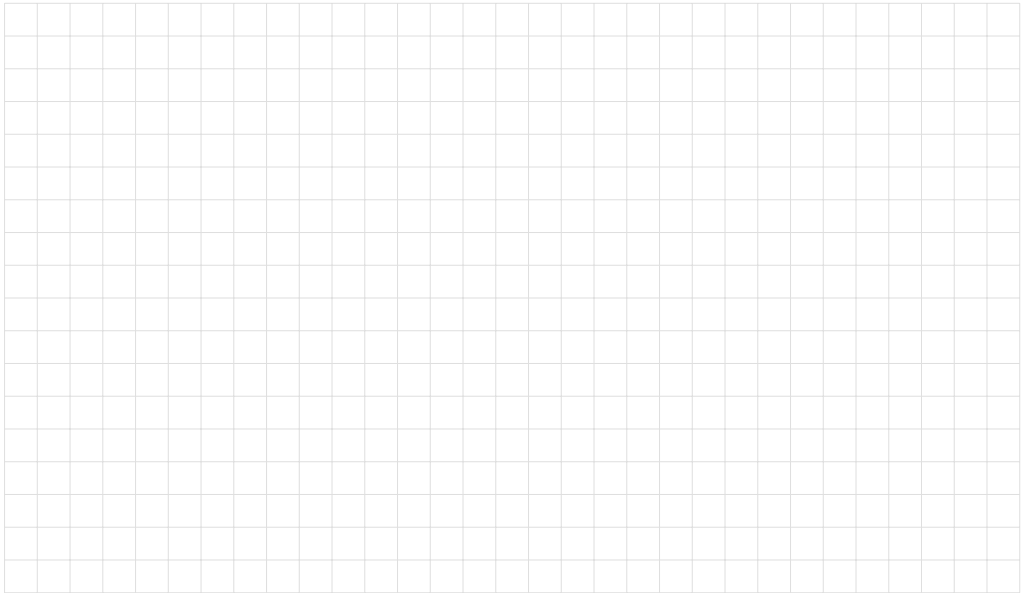
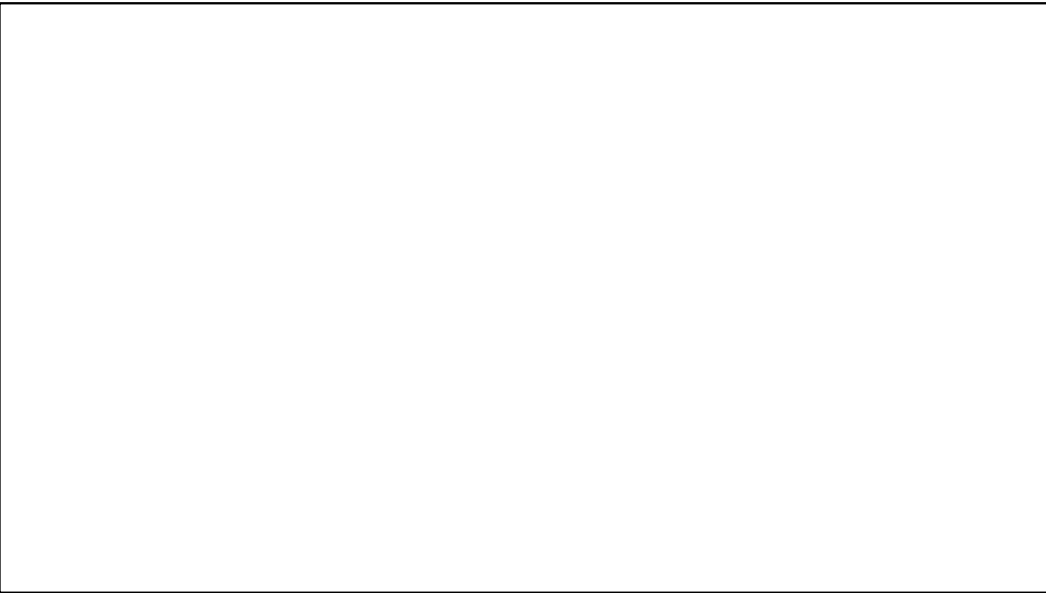
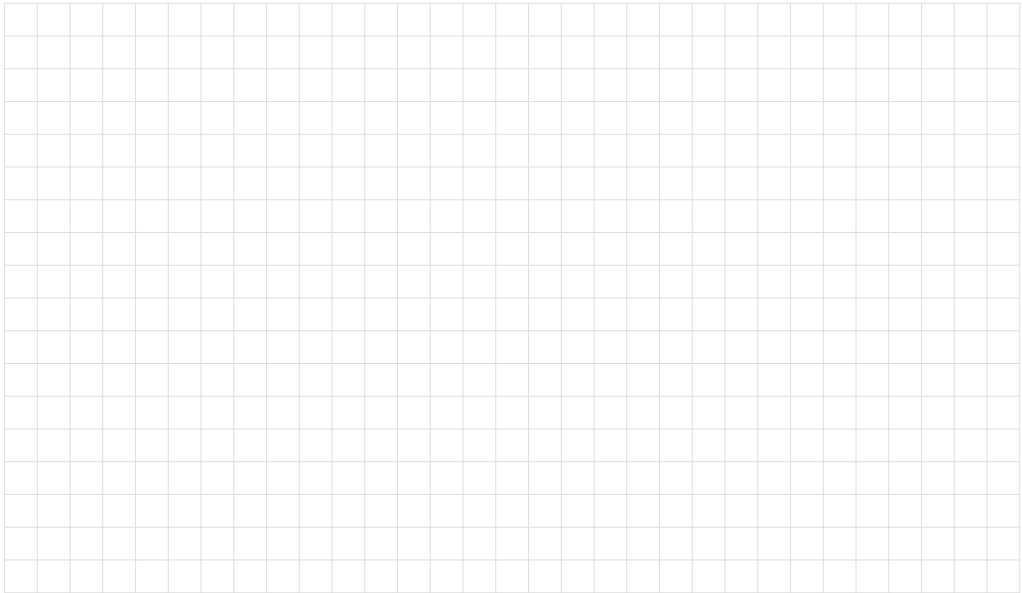
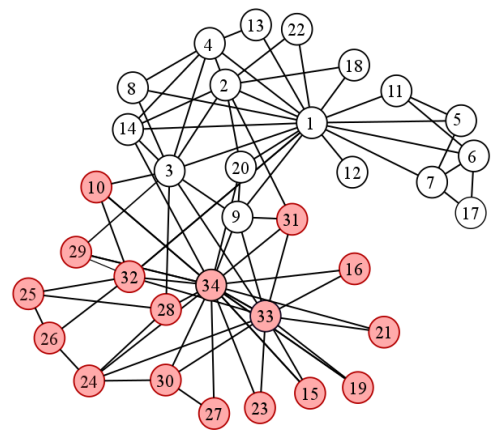
$$a_{51} = 1.5$$

Food for thought

The graph shown in the next slide is known as the *Zachary Karate Club Graph*: nodes represent club members and an edge indicates a social interaction/friendship outside the club activities.

1. Is the graph connected?
2. Within each color group (red/white), find the nodes with largest and lowest degree.
3. Consider a directed version obtained by orienting each edge.
 - Is it possible to orient the edges so that the digraph is strongly connected?
 - Is it possible to orient them so that there is exactly one globally reachable node?

Food for thought

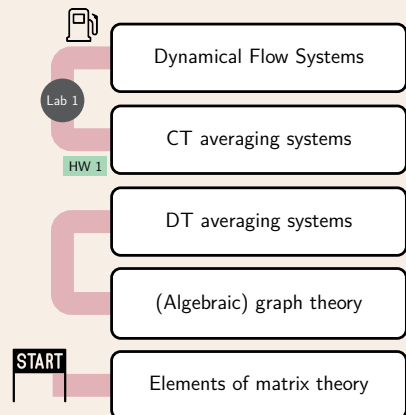


Elements of Algebraic Graph Theory

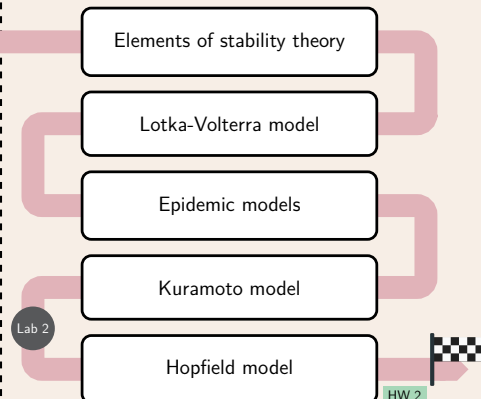
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Nonlinear Dynamics



Covered topics

- ▷ Adjacency matrix
- ▷ Graph theoretical characterization of irreducible matrices
- ▷ Graph theoretical characterization of primitive matrices
- ▷ Applications to DT averaging systems



The adjacency matrix (of a weighted digraph)

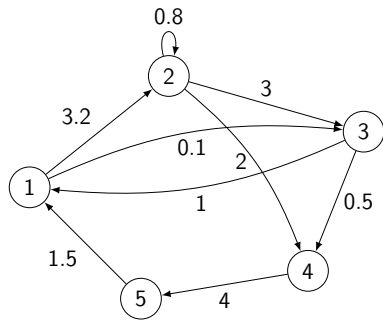
Def: Given a weighted digraph $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ of order n the (weighted) adjacency matrix of G is the $n \times n$ matrix A in which:

- (i) for each edge $(i, j) \in E$ the (i, j) entry of A is equal to the edge weight a_{ij} ,
- (ii) all other entries of A are equal to zero.

- ▷ We will consider weighted digraphs with positive weights $\implies A \geq 0$
- ▷ The binary adjacency matrix $A_{0,1}$ of G is the “binarized” $(0,1)$ of A
- ▷ The weighted out-degree and in-degree matrix of G are defined as

$$D_{\text{out}} = \text{diag}(A \mathbb{1}_n) = \begin{bmatrix} d_{\text{out}}(1) & & \\ & \ddots & \\ & & d_{\text{out}}(n) \end{bmatrix}, \quad D_{\text{in}} = \text{diag}(A^T \mathbb{1}_n) = \begin{bmatrix} d_{\text{in}}(1) & & \\ & \ddots & \\ & & d_{\text{in}}(n) \end{bmatrix}$$

The adjacency matrix (of a weighted digraph)



$$A = \begin{bmatrix} 0 & 3.2 & 0.1 & 0 & 0 \\ 0 & 0.8 & 3 & 2 & 0 \\ 1 & 0 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 1.5 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The adjacency matrix (of a weighted digraph)

Q: Fill in the blanks

G undirected $\Rightarrow d_{ij} = d_{ji} \Rightarrow A = A^T$

G weight-balanced $\Rightarrow d_{\text{out}}(v) = d_{\text{in}}(v) \quad \forall v \in V \Rightarrow D_{\text{in}} = D_{\text{out}}$

$i \in V$ is a sink $\Rightarrow d_{\text{out}}(i) = 0 \Rightarrow a_{ij} = 0 \quad \forall j \Rightarrow i\text{-th row of } A = 0$

$i \in V$ is a source $\Rightarrow d_{\text{in}}(i) = 0 \Rightarrow a_{ji} = 0 \quad \forall j \Rightarrow i\text{-th col of } A = 0$

G has no self-loops $\Rightarrow \text{diagonal of } A = 0$

$D_{\text{out}} = I \Rightarrow A \mathbf{1}_n = \mathbf{1}_n \Rightarrow A \text{ row-stochastic}$

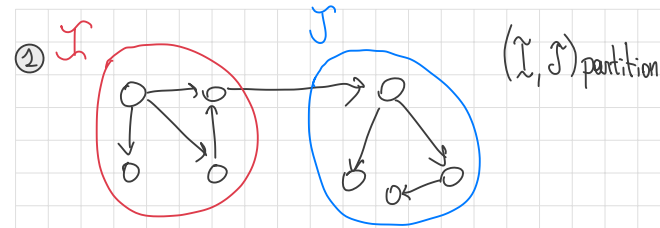
$D_{\text{out}} = D_{\text{in}} = I \Rightarrow A \mathbf{1}_n = \mathbf{1}_n, \mathbf{1}_n^T A = \mathbf{1}_n^T \Rightarrow \text{stoch}^2$

Graph theoretical characterization of irreducible matrices

- ① $\triangleright \{\mathcal{I}, \mathcal{J}\}$ is a partition of $V = \{1, \dots, n\}$ if $\mathcal{I}, \mathcal{J} \neq \emptyset$, $\mathcal{I} \cap \mathcal{J} = \emptyset$, $\mathcal{I} \cup \mathcal{J} = V$
- ② $\triangleright P \in \{0, 1\}^{n \times n}$ is a permutation matrix if it has exactly one entry equal to 1 in every row and every column

Lemma (Walks and powers of A): Let G be a weighted digraph with adjacency matrix A and binary adjacency matrix $A_{0,1}$. Then, for all $k \in \mathbb{N}$:

- (i) the (i, j) entry of $A_{0,1}^k$ equals the number of directed walks of length k from node i to node j ,
- (ii) the (i, j) entry of A^k is positive $\iff$ there exists a directed walk of length k from node i to node j .



② $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ permutation matrix

Note 1) The cols. of permutation matrix P are canonical basis vectors

2) If P is a permutation $P^T P = I_n \Rightarrow P^{-1} = P^T$

3) If A is the adjacency matrix of a weighted digraph and P is a permutation matrix, then PAP^T is the adjacency matrix of G up to a re-labeling of nodes

Example $\begin{matrix} 2 & 1 & 3 \\ \text{---} & \text{---} & \text{---} \\ 3 & 2 & 1 \end{matrix} \xrightarrow{(2,1,3) \rightarrow (3,2,1)} \begin{matrix} 3 & 2 & 1 \\ \text{---} & \text{---} & \text{---} \\ 1 & 2 & 3 \end{matrix}$ $P = \begin{bmatrix} e_{\pi(1)} & e_{\pi(2)} & e_{\pi(3)} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \xrightarrow{P} \bar{A} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Proof (only of (ii))

We prove (ii) by induction on k

Base case: ($k=1$) If $k=1$ $[A^k]_{ij} > 0 \Leftrightarrow j \in N^{\text{out}}(i) \Leftrightarrow \exists$ walk of length 1 from i to j

Induction step: ($k \Rightarrow k+1$) Assume that the statement holds for a generic k , that is:

$$[A^k]_{ij} > 0 \Leftrightarrow \exists \text{ walk of length } k \text{ from } i \text{ to } j$$

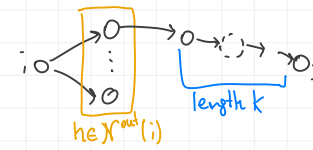
We want to prove that this holds for $k+1$. Note that $A^{k+1} = A \cdot A^k$, then

$$[A^{k+1}]_{ij} = [A \cdot A^k]_{ij} = \sum_{h=1}^n [A]_{ih} [A^k]_{hj}$$

$\Downarrow$ $\Downarrow$
 $h \in N^{\text{out}}(i)$ $\exists$ walk $h \rightarrow j$ of length k

So $[A^{k+1}]_{ij} > 0 \Leftrightarrow \exists$ walk of length k from $\text{hex}^{\text{out}}(i)$ to j

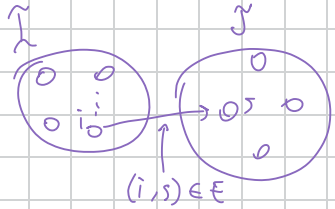
$\Leftrightarrow \exists$ walk of length $k+1$ from i to j



Graph theoretical characterization of irreducible matrices

Thm (Irreducible matrices and digraphs): Let $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ be a weighted digraph of order $n \geq 2$ with adjacency matrix A . The following statements are equivalent:

- (i) A is irreducible
- (ii) there exists no permutation matrix P such that PAP^T is block upper triangular
- (iii) G is strongly connected
- (iv) for all partitions $\{I, J\}$ of V , there exists $i \in I$ and $j \in J$ such that $(i, j) \in E$



$\exists r \geq 2 \leq A$
 $PAP^T = \begin{bmatrix} B & C \\ 0 & D \end{bmatrix}$ B is $r \times r$
 D is $(n-r) \times (n-r)$

Proof: (of some selected statements)

- "(i) $\Rightarrow$ (iii)" Assume A is irr. Then, by def, $\sum_{k=0}^{n-1} A^k > 0$. This implies that $\forall i, j \in \{1, \dots, n\} \ i \neq j \ \exists k_{ij} \geq 2$ st $[A^{k_{ij}}]_{ij} > 0$

By the previous lemma, this in turn implies that $\forall i, j \in V \ i \neq j \ \exists$ walk of length k_{ij} from i to j . Thus, G is strongly connected

- "(iii) $\Rightarrow$ (i)" Assume that G is strongly connected. Then $\forall i, j \in V \ i \neq j \ \exists$ a walk of length k_{ij} from i to j . By the previous lemma this implies that $[A^{k_{ij}}]_{ij} > 0$

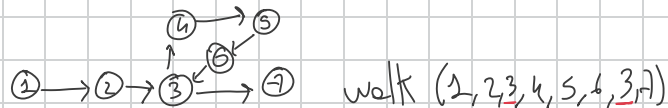
To prove that $\sum_{k=0}^{n-1} A^k > 0$ (ie: A is irreducible), it remains to show that it is always possible to choose $k_{ij} \leq n-1$

To this end, consider an arbitrary walk of length k_{ij} from i to j :

1) If $k_{ij} \leq n-1$ we are done

2) If $k_{ij} > n-1$, then the walk must contain some repeated nodes. Note that the sub-walk between any two repeated nodes form a cycle.

Example:

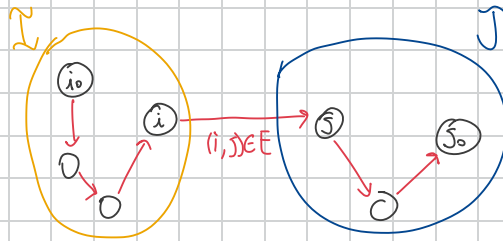


By removing all cycles in my walk we obtain a walk with no repeated nodes of length $\leq n-1$

• "(iii) $\Rightarrow$ (iv)"

We assume that G is strongly connected. Pick a partition $\{\mathcal{I}, \mathcal{J}\}$ of $V = \{1, \dots, n\}$ and two nodes $i_0 \in \mathcal{I}$ and $j_0 \in \mathcal{J}$

Since G is strongly connected, $\exists$ a walk from i_0 to j_0 . Hence, there must exist an edge from a node in $\mathcal{I}$ to a node in $\mathcal{J}$



• "(iv) $\Rightarrow$ (iii)"

Pick a node $i \in V$. Let R_i be the set of nodes that can be reached from i through a walk and let U_i be the set of nodes that cannot be reached from i through a walk. Note that $R_i \cup U_i = V$

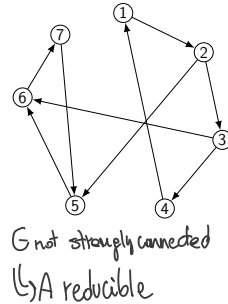
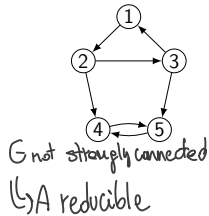
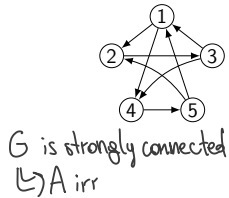
By contradiction that $U_i \neq \emptyset$. This implies that G is not strongly connected. Then $\{R_i, U_i\}$ is a partition of V and (iv) implies the existence of an edge (j, h) , $j \in R_i$, $h \in U_i$.

But then this implies that h is reachable from i . But then node $h \in U_i$ is reachable from $i \Rightarrow$ CONTRADICTION

Hence it must be that $U_i = \emptyset$ and all nodes in V are reachable from i . Since i is arbitrary, G must be strongly connected

Graph theoretical characterization of irreducible matrices

Q: The adjacency matrix of the following digraphs is irreducible?



$$A_2 = \begin{bmatrix} 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 2 & 0 \end{bmatrix}$$

block upper triangular matrix

Graph theoretical characterization of primitive matrices

Lemma (Frobenius number): Given a finite set $A = \{a_1, \dots, a_n\}$, $a_i \in \mathbb{N}$, an integer M is said to be representable by A if there exist $\{\alpha_1, \dots, \alpha_n\}$, $\alpha_i \in \mathbb{N}$, such that $M = \alpha_1 a_1 + \dots + \alpha_n a_n$. The following statements are equivalent:

- (i) there exists a (finite) largest unrepresentable integer (Frobenius number of A),
- (ii) the greatest common divisor of A is 1. (they are coprime)

Def: Given a digraph $G = (V, E)$ the recurrence time of $i \in V$ is an integer $k \geq 1$ such that there exists a directed walk from i to i of length k . The period of i is the greatest common divisor of all recurrence times of i .

Lemma (Strongly connected aperiodic digraphs): If G is strongly connected, then G is aperiodic if and only if the period of each node is 1.

Example (to explain Frobenius number)

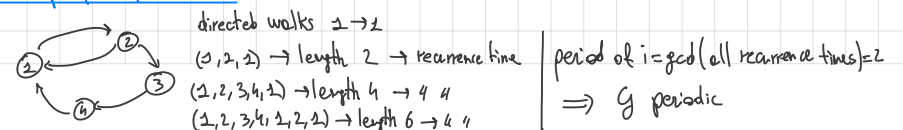
1) $A = \{7, 12\}$, $\gcd(7, 12) = 1 \Rightarrow \exists$ finite largest unrepresentable integer which is 65

2) $A = \{5, 10\}$, $\gcd(5, 10) = 5 \Rightarrow$ infinite unrepresentable numbers

Coin problem: Given coins of different values, which is the largest monetary amount that can be obtained using those coins?

Ex: Coins of 2, 5 cents $\rightarrow$ Frobenius number = 3
 Any amount > 3 (obviously integer) can be paid with coins of 2 and 5 cents
 4 cents = $2 \cdot 2$ cents
 12 cents = $2 \cdot 5$ cents + 2 cents = $6 \cdot 2$ cents

Example (to explain lemma)



Graph theoretical characterization of primitive matrices

Thm (Primitive matrices and digraphs): Let $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ be a weighted digraph of order $n \geq 2$ with adjacency matrix A . The following statements are equivalent:

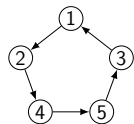
- (i) A is primitive,
- (ii) G is strongly connected and aperiodic. $\gcd(\text{lengths of all simple cycles}) = 1$

Corollary (Primitive matrices and digraphs with self-loops): Let $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ be a weighted digraph of order $n \geq 2$ with adjacency matrix A . If G has at least one self-loop, and is strongly connected, then A is primitive.

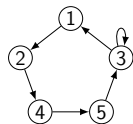
$\hookrightarrow G$ aperiodic

Graph theoretical characterization of primitive matrices

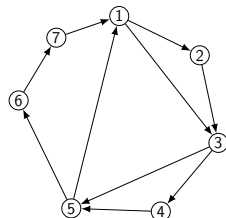
Q: Is the adjacency matrix of the following digraphs primitive?



cycle $c_1 = (2, 2, 4, 5, 1, 2)$
length $\ell_1 = 5$
 G is periodic
 $\Rightarrow A$ not primitive



A primitive



$\gcd(\ell_1, \ell_2) = 1 \Rightarrow G$ aperiodic
 A primitive

Thm (Primitive matrices and digraphs): Let $G = (V, E, \{a_{ij}\}_{(i,j) \in E})$ be a weighted digraph of order $n \geq 2$ with adjacency matrix A . The following statements are equivalent:

- (i) A is primitive,
- (ii) G is strongly connected and aperiodic.

Proof:

"(i) $\Rightarrow$ (ii)" Assume A primitive, $\exists k \in \mathbb{N}$ st. $A^k > 0$. From the Lemma on walks & powers of A , we know that $\forall i, j \in V \exists$ walk of length k from i to j . Then G is strongly connected.

To prove aperiodicity, note that since G is strongly connected, then each node $i \in V$ has at least one out-neighbor. That is, $\forall i \in V \exists j \in V$ st. $a_{ij} > 0$.

$$\text{Hence } \forall i \in V: \underbrace{[A^{k+2}]_{ii}}_{A \cdot A^k} = \sum_{h=2}^n a_{ih} [A^k]_{hi} \geq \underbrace{a_{ij}}_{>0} \underbrace{[A^k]_{ji}}_{>0} > 0$$

This implies that $A^{k+2} > 0$. Therefore, $\forall i \in V$, there are walks from i to i of lengths k and $k+2$. Hence, since $\gcd(k, k+2) = 2$, the period of all nodes in V is 2.

By the previous lemma, G is aperiodic.

"(ii) $\Rightarrow$ (i)" Assume G strongly connected. Let $\{C_1, \dots, C_N\}$ be the simple cycles of G and $\{l_1, \dots, l_N\}$ the corresponding lengths.

Pick $i, j \in V$. Since G is strongly connected, $\exists$ walk of length L_{ij} from i to j so that the walk visits at least one node of each simple cycle.

By attaching to this walk:

$$\left\{ \begin{array}{l} \text{cycle } C_1 \text{ repeated } d_1 \in \mathbb{N} \text{ times} \\ \text{cycle } C_2 \text{ repeated } d_2 \in \mathbb{N} \text{ times} \\ \vdots \\ \text{cycle } C_N \text{ repeated } d_N \in \mathbb{N} \text{ times} \end{array} \right.$$

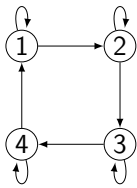
By doing so, we obtain a walk from i to j of length: $L_{ij} + \alpha_1 l_1 + \alpha_2 l_2 + \dots + \alpha_n l_n$

Since G is aperiodic then $\gcd(l_1, \dots, l_n) = 1$ and by Frobenius number lemma $\exists \bar{L}$ st $\forall M > \bar{L}$, $M = \alpha_1 l_1 + \dots + \alpha_n l_n$ for some integer coefficients α_i .

Hence we can find walks from i to j of any lengths $\geq L_{ij} + \bar{L} + 1$. Equivalently, $[A^k]_{ij} > 0 \quad \forall k \geq L_{ij} + \bar{L} + 1$

By applying the same argument to all pairs $i, j \in \bar{V}$ we conclude that $\exists$ sufficiently large k st $A^k > 0$, that is A is primitive.

Applications to DT averaging systems



$$x(k+1) = Ax(k)$$

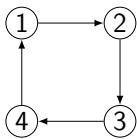
$$A = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0 & 0.5 \end{bmatrix}, x(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \end{bmatrix}, x(0) = \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ x_{4,0} \end{bmatrix}$$

$$A \text{ doubly-stochastic} + \text{primitive} \implies w = \frac{1}{4}\mathbb{1}_4 \implies \lim_{k \rightarrow \infty} A^k = \frac{1}{4}\mathbb{1}_4\mathbb{1}_4^T$$

$$\lim_{k \rightarrow \infty} x(k) = \lim_{k \rightarrow \infty} A^k x(0) = \frac{1}{4}\mathbb{1}_4\mathbb{1}_4^T x(0) = \left(\frac{1}{4} \sum_{i=1}^4 x_{i,0} \right) \mathbb{1}_4$$

average of initial conditions } for every node

Applications to DT averaging systems



$$x(k+1) = Ax(k)$$

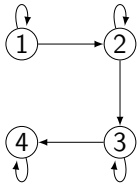
$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}, x(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \end{bmatrix}, x(0) = \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ x_{4,0} \end{bmatrix}$$

Cyclic shift

$$A \text{ row-stochastic} + \text{irreducible but NOT primitive} \implies \nexists \lim_{k \rightarrow \infty} x(k)$$

$$x(2) = Ax(0) = \begin{bmatrix} x_{2,0} \\ x_{3,0} \\ x_{4,0} \\ x_{1,0} \end{bmatrix} \quad x(2) = Ax(2) = \begin{bmatrix} x_{3,0} \\ x_{4,0} \\ x_{1,0} \\ x_{2,0} \end{bmatrix}$$

Applications to DT averaging systems



$$x(k+1) = Ax(k)$$

$$A = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}, x(k) = \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \end{bmatrix}, x(0) = \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ x_{4,0} \end{bmatrix}$$

A row-stochastic + NOT irreducible $\Rightarrow \lim_{k \rightarrow \infty} x(k) = ?$

A has a simple and strictly dominant eigenvalue 1

$A^k \xrightarrow{k \rightarrow \infty} w^T \mathbf{1}$ where w is the left dominant eigenvector of A

What is the form of w ?

$$w^T A = w^T \quad w^T = e_i^T \Rightarrow e_i^T A = e_i^T$$

$$x(k) \xrightarrow{k \rightarrow \infty} A^k x(0) = \mathbf{1}_n e_i^T x(0) = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{1,0} \\ x_{2,0} \\ x_{3,0} \\ x_{4,0} \end{bmatrix} = x_{4,0} \mathbf{1}_4$$

Node 4 is the globally reachable node in the graph